

Research Assignment 1

Section A:

- Q1. Relational Databases: Store data in tables
- NoSQL Databases: Uses flexible data models & Hierarchical database - Store data
- Hierarchical Database: Organize data in a tree-like structure, where each parent record can have multiple child records.
- Network Database: Build on hierarchical model but allow child records to be linked to multiple parent records.
- Object oriented Databases: Based on the principles of OOP where data is stored as objects.
- Cloud Databases: Operates in a cloud environment hosted on cloud computing platforms.
- Centralised Databases: A database stored and managed at a single location
- Personal Databases: A small-scale database designed for a single user.
- Operational Databases: Designed to manage and process real-time data for daily operations within organization and businesses.

Q2 A Relational Database Management System is software that allows users to create, manage, and interact with relational databases which store data in structured tables of rows and columns.

Q3 - A primary key is a unique identifier for each record in a table, ensuring that no two rows have the same value and cannot contain ("Null").

A Foreign key is a column in one table that links to the primary key of another table, establishing a relationship between two tables.

Q4 Data normalization is the process of organizing data to reduce redundancy and improve data integrity.

Q5 A data schema is a blueprint that defines how data is organized within a database, this includes the structure of tables, fields and the relationships between them.

Q6

Structured data	Semi-structured	unstructured
- Fixed Format	- some properties	- lacks structure
- less flexible	- are organized	- Versioned as a
- very robust	- New technology	Whole

Q7 Fact-table: They store quantitative data related to the business process (numeric)
Dimension table: Provide Context for the data in the Fact table

Q8 A data model is a Formal Structure that defines how raw interaction data is organised, stored and interpreted to generate actionable insights. A well designed data model ensures consistency across reporting.

Q9 Database is used to store structured data and supporting transactions.
A Datawarehouse is designed for analytical processing, storing and structured data ~~for~~ From various sources.
A Data LAKE store large volumes of data in their original format without a predefined Schemas.

Q10 **Data Mart** is a smaller specialized Subset. The difference between the Data Mart and data Warehouse is that the mart is used for smaller specialized Subject area while data warehouse integrates data from multiple sources across the entire organization.

Section B

Q11 A query language is a specialized ~~computer~~ ^{computer} language used to retrieve and manipulate data from databases. It is used widely due to its simple process of accessing and managing data.

Q12 Indexing in DBMS is used to speed up data retrieval by minimizing disk scans.

Q14 The core ~~soft~~ software component that manages how data is created, read, updated and deleted in a database.

It impacts performance by determining how efficiently data is stored ^{and} accessed with features like indexing and transaction support playing crucial roles in optimizing query speed and resource usage.

Q13 Transactions are sequence of operations that are treated as a single unit of work, ensuring that either all operations are completed successfully or none are applied.

~~ACID in a database~~

ACID properties: Atomicity, Consistency, Isolation, and Durability.

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Q5 Views are virtual tables created by querying data from one or more tables, allowing users to simplify complex queries.

Stored procedures are reusable SQL code that can perform various tasks and parameters.

Triggers are special stored procedures that automatically execute in response to specific events on a table (INSERT, UPDATE, DELETE).

Q16	ELT	ETL
	- Traditional technique of extracting raw data	- Extract, load, Transform
	Extract load	- Faster processing speed
	- Extract transform load	- Compatible with data lakes
	- Slower processing speed	
	- Primary structured data	

Q17 Batch Processing involves collecting large volumes of data and processing them in groups.

Stream processing: stream processing continuously ingests and analyses data in real time as it flows through the system.

Batch Processing	Stream Processing
- Scheduled in bulk	- Continuous in real time
- High latency	- Low latency

A join is a clause that is used to combine rows from two or more tables based on a related column.

Inner JOIN - Returns record that have matching values in both tables

LEFT JOIN - Returns all records from the left table, and the matched records from the right table

Right JOIN - Returns all the records from the right table, and the matched records from the left table.

Full JOIN - Returns all records when there is a match in either left or right table.

Q19 Referential integrity is a concept in the database management that ensures all data references between tables are valid and consistent.

Q20 Data inconsistency caused by data redundancy leads to slower decision making. Performance is slowed down. Storage gets affected. ~~Return~~ Redundancy causes the request of additional storage to store data.

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Section C:

- Q20 ~~Cloud databases are hosted~~
- Q21 Cloud databases are hosted on third-party servers and offer easy scalability and lower upfront cost (less infrastructure to worry about).
On-premises are managed on local servers which provide greater control and security.
- Q22 According to IBM, data governance is the data management discipline that focuses on the quality, security and availability of organization's data. Its important because it provides us with the ability to ensure data integrity and data security by defining and implementing policies, standard and procedures for data collection, ownership, storage, processing and use.
- Q23 Data integrity refers to the accuracy, completeness, and consistency of data throughout its lifecycle.
It is maintained maintained through practices such as data validation, access control, encryption, regular backups and audit trails.

Q24 Data quality refers to the accuracy, completeness, consistency, ~~timeliness~~ timeliness of data ensuring it fits intended purpose.

It's important in terms of data analytics because high-quality data leads to reliable data (insights) and informed decision-making.

Q25 According to the Northeastern University The key responsibilities (Designing and maintaining data systems and databases such as fixing coding errors This falls in the data management. Data collection and cleaning to ensure the source of information is valid and complete.

Q26 According to COMPTIA the responsibilities of a DB admin are as follows:

- updating databases (generating new permissions for database access)
- organizing and maintaining data
- Designing and developing databases
- Archiving data - data that is used less is removed.
- Maintaining and troubleshooting databases
- Improving database security
- Creating database documentation

- training others in database management
- Performing backup and recovery
- Planning for future growth

Q27 Main steps in data pipeline

- Define pipeline's goal and architecture
- Selecting the data sources and the ingestion strategies
- Processing and transforming the data and storing in a destination
- Setting up monitoring and maintenance for ongoing operations.

Q28 Challenges presented by managing large scale databases are as follows:

Index - Identify Root causes of lock Contention

- Investing in Scalable Hardware solutions
- Implementing Robust Backup Strategies
- Optimize Slow Query Performance
- Mitigate Security Vulnerabilities Proactively
- Ensure Database Health with Effective monitoring

Q29 PL/SQL - Procedural language for SQL is an extension of SQL in Oracle databases, best used for Enterprise Resource Planning systems (banking, telecoms)

- Microsoft SQL Server
Is developed by Microsoft used for
E-commerce platforms

- MySQL (wordpress)
open source relational database
management system that uses
SQL for querying and managing
data. used for startup business
like using Wordpress (we have
used it at some point)

Q30 Top most used storage formats

- 1- CSV Comma-Separated values
its compatible
- 2- JSON - Javascript object Notation
supports complex data structures
- 3- Parquet is optimized for performance
and storage efficiency in
data environments.